Evaluation & Study Design

Day 3 — How would you know if it worked?

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Where we are on the spine

Handoff — participants recap their Day 2 data specification.

- **Yesterday:** a problem → the data that exists to attack it.
- **Today:** data → a **credible way to know** whether a solution works.

Day 2: provenance & who is missing

The central question

Before you build anything:

What would count as success?

A method you **cannot evaluate credibly** is not yet a contribution.

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The validity ladder

"Evaluation" otherwise smears across three days. Name the rungs:

- **Internal validity** — is the model sound on its own terms? *(Days 3–4)*
- **External / deployment validity** — does it hold in the world? *(Day 5)*

Today is the bottom rung: study design.

FIG F3.1 **The validity ladder.** A two/three-rung ladder: study design → internal validity (Days 3–4) → external/deployment validity (Day 5). Reused on Day 4 and Day 5 to show the rung change. *See build guide F3.1.*

Two things both called "evaluation"

- **(a) Study design** — defining the *target* and how you would credibly *measure* it. ← **today.**
- **(b) Evaluating a model you have built** — Days 4 and 5.

Today is **(a):** you design the evaluation **before the model exists.**

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NAMING THE TARGET

Defining what you are actually measuring

Estimands — what are you actually measuring?

- Name the **quantity of interest** precisely — *before* choosing a metric.
- A vague target produces an **unfalsifiable claim**: anything can look like success.

Vague (not an estimand)

"Improve sepsis prediction."

Precise (an estimand)

"Sensitivity at a 90%-specificity threshold, in adult ICU admissions, for sepsis onset within 6 hours."

The anatomy of an estimand

Every credible target names four things:

ELEMENT	THE SEPSIS EXAMPLE
Population	adult ICU admissions
Outcome	sepsis onset
Timing / horizon	within 6 hours
Operating point	sensitivity at 90% specificity

Write these down before you touch a model. If you can't, you don't yet have an evaluable claim.

Target population vs. deployment population

- The population you **evaluate on** is rarely the population you **deploy on.**
- The gap is a **study-design decision**, not an afterthought.
- Who is in the evaluation set is inherited from the data's provenance.

FIG F3.3 **Eval vs. deploy population.** Two partly-overlapping distributions (e.g., age or acuity) labeled "evaluation cohort" and "deployment population," with the non-overlap shaded as the generalization gap. *See build guide F3.3.*

Day 2: who is missing

CHOOSING WHAT TO MEASURE

Metrics

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Metrics: utility vs. convenience

- **Metrics of convenience** — AUROC for everything, because it is easy.
- **Metrics of utility** — chosen because they track the **clinical decision.**

TOOL	TELLS YOU	DOES *NOT* TELL YOU
Discrimination (AUROC)	Ranking ability	Whether probabilities are usable
Calibration	Are the probabilities honest	How well it ranks
Decision curves	Net benefit at real thresholds	Anything outside that range

Choose the metric the clinical decision implies.

Discrimination — and its limits

- **AUROC** — probability a random positive outranks a random negative; prevalence-independent.
- **AUPRC** — area under the precision–recall curve; prevalence-dependent.
- Both are **threshold-free summaries**: they rank models but say nothing about whether the *probabilities* are usable, or whether any single operating point helps a decision.

Discrimination is necessary, not sufficient. Calibration and decision utility come next.

At the operating point, prevalence still bites

A sepsis screen, **1% prevalence**, AUROC = 0.90, threshold set for **80% sensitivity**:

- At a plausible specificity, **most positive alarms are false** — PPV can sit in the single digits.
- The clinician experiences **alarm fatigue**, not a 0.90 model.
- A strong threshold-free summary can coexist with a useless **operating point.**

→ Day 5: alarm fatigue

Calibration — necessary, not sufficient

- Discrimination and calibration are **independent**: a model can rank perfectly and still output dishonest probabilities.
- **Global calibration is insufficient** — check **subgroup** reliability, and where warranted **multicalibration** (Hébert-Johnson et al., 2018); pick the target from the decision and the clinically meaningful strata.
- Calibration **decays under shift** — so it is a *deployment-time* property needing recalibration, not a one-time check.

FIG F3.5 **Reliability curve — overall vs. by subgroup.** The diagonal, the overall curve near it, and a subgroup curve that deviates sharply — the case aggregate calibration conceals. *See build guide F3.5.*

Clinical utility — net benefit, and whose threshold

- **Decision-curve analysis** — does using the model beat "treat all" / "treat none"? Net benefit sweeps a **range of threshold probabilities** and folds in the **relative cost** of FP vs. FN.
- The open question is **whose** threshold matters — patient, clinician, institution, or payer — and how a system **operationalizes** one.
- Heterogeneous utilities may call for **stratified or individualized** threshold policies, not a single cohort number.

FIG F3.6 **Decision curve.** Net benefit (y) vs. threshold probability (x), model curve above "treat all"/"treat none" over the relevant threshold range. *See build guide F3.6.*

What aggregate numbers hide

Fairness and disaggregated evaluation

- **Aggregate performance can hide subgroup failure** — a strong overall number can mask a model that fails an entire population.
- **Disaggregated evaluation** is a default, not an add-on.
- Decide the subgroups, and the per-subgroup targets, *as part of the design.*

FIG F3.7 **Aggregate hides subgroups.** One reassuring overall bar beside a panel of per-subgroup bars, one of which is clearly failing. *See build guide F3.7.*

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Power, multiplicity, and the winner's curse

- Underpowered studies produce **inflated, unreplicable positives**, not just misses (Card et al., "With Little Power").
- ML adds **multiplicity**: searching over models, seeds, and thresholds is uncontrolled multiple testing — the reported number is the **max of a search**.
- And the test set **leaks** — reuse, tuning to it, and train–test **contamination** (acute for LLMs) inflate everything downstream.

> Your headline metric is a *selected* statistic. Treat it like one.

The medical algorithmic audit

A **structured framework** for evaluating a clinical model end to end (Liu et al., 2022) — a named thing participants can apply.

- Specify the intended use, then probe systematically for failure.
- Test subgroups, edge cases, and the points where the model meets the workflow.
- An auditable model is one whose failures **don't hide.**

This is **algorithmic safety**, not box-checking: **observational causal inference** lets you audit *and improve* a model — asking what it would do under intervention, not just how it scored.

Evidentiary standards — the bar for a real claim

- **RCTs of ML interventions** (Plana et al., 2022) — the high bar for a *deployed* claim; few clear it.
- **Effect heterogeneity** (Yu et al., 2024) — an average effect can hide who is helped and who is harmed.
- "Works on average" ≠ "shown to help patients."

Threats to internal validity

The quiet study-killers — name them in advance:

- **Leakage** — future or label information in the features.
- **Shortcut learning** — the model solves a predictive **proxy**, not the task; a named, general phenomenon (Geirhos et al., 2020), not a quirk.
- **Anticausal pitfalls** — predicting cause-from-effect couples to the label-generating process, so shortcuts hide easily (Schölkopf et al., 2012).

These invalidate a study *silently* — and a clean held-out split does **not** catch them.

The canonical case — COVID-CXR shortcuts

How a "diagnostic" model learned everything but the disease:

- Deep models flagged COVID from chest X-rays with high reported accuracy...
- ...by reading **scanner make, patient position, and dataset source** — not pathology (DeGrave et al., 2021).
- It generalized within the benchmark and **collapsed** off it. The held-out split shared the shortcut.

The tell: a feature (or artifact) that wouldn't exist, or wouldn't be informative, at the real decision point.

Frontier — evaluating without ground truth

- Often there are **no timely labels** (live deployment) or **no single right answer** (generative output).
- **Selective prediction / abstention** — let the model decline; report the **risk–coverage** tradeoff, not one accuracy.
- **Conformal prediction** — distribution-free prediction *sets* with finite-sample coverage guarantees (Angelopoulos & Bates, 2021).
- **LLM-as-judge** — convenient and **systematically biased**; an evaluation artifact, not ground truth.

> With no ground truth, "what's the accuracy?" is the wrong question — calibrated uncertainty and abstention are the right ones.

→ Day 4: uncertainty at modeling time

→ Day 5: monitoring without labels

The afternoon — build your own evaluation

The 2:45 notebook runs end to end on your project — **Sections 0–7:**

- **0** Project setup → **1** Discrimination → **2** Calibration (overall + subgroup)
- **3** Clinical utility (decision curves) → **4** Disaggregated audit → **5** Uncertainty & power
- **6** Leakage & shortcut probe → **7** Findings → Workbook Part 3

Worked example, template, and AI-coding-tool guidance are provided — see the Day 3 & 4 Workshops doc.

Dissect, then design

Today's deliverable — Evaluation & study-design plan.
Your **estimand**, the **metrics** the clinical decision implies, your **subgroup/fairness** evaluation, and the top **threats to validity** for your project.

Next: 11:00 small group — dissect a published evaluation, *does this evidence convince you?* · 1:30 guest (Olawale Salaudeen) — disaggregated metrics · 2:45 evaluation notebook.